

LAB ASSIGNMENT – 2

1. Name: ~~Arpita Saha~~

Roll No: UG/SOET/30/24/398

DATE: 19/01/2024

Q1. Make a schema first and then insert 6 documents.

- Roll_no=[1,2,3,4,5,]
- Name=["Ram", "Alex", "John", "Bob", "Mukesh", "Danny"]
- Age=[20,19,40,55,30,28]
- Salary=[546.7,333.4,666.7,678.4,245.6,546.3]
- City=["A", "B", "C", "D", "E", "F"]
- Phone_no=[123,456,122,444,567,892]

```
Query: db.students.insertMany([
... {roll_no: 1, name: "Ram", age: 20, salary: 546.7, city: "A", phone_no: 123},
... {roll_no: 2, name: "Alex", age: 19, salary: 333.4, city: "B", phone_no: 456},
... {roll_no: 3, name: "John", age: 40, salary: 666.7, city: "C", phone_no: 122},
... {roll_no: 4, name: "Bob", age: 55, salary: 678.4, city: "D", phone_no: 444},
... {roll_no: 5, name: "Mukesh", age: 30, salary: 245.6, city: "E", phone_no: 567},
... {roll_no: 6, name: "Danny", age: 28, salary: 546.3, city: "F", phone_no: 892}
...])
```

```
Output: {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('697611febf3c95f1a01e2621'),
    '1': ObjectId('697611febf3c95f1a01e2622'),
    '2': ObjectId('697611febf3c95f1a01e2623'),
    '3': ObjectId('697611febf3c95f1a01e2624'),
    '4': ObjectId('697611febf3c95f1a01e2625'),
    '5': ObjectId('697611febf3c95f1a01e2626')
  }
}
```

Q2. Write a query to update the name of RAM to SAM. Query:

```
db.students.updateOne(  
  ... { name: "Ram" },  
  ... { $set: { name: "Sam" } }  
  ... )
```

Output: {
 acknowledged: true,
 insertedId: null,
 matchedCount: 1,
 modifiedCount: 1,
 upsertedCount: 0
}

Q3. Write a query to display only the cities present in that collection. Query:

```
db.students.find({}, { city: 1, _id: 0 })
```

Output: [{ city: 'A' },
 { city: 'B' },
 { city: 'C' },
 { city: 'D' },
 { city: 'E' },
 { city: 'F' }]

Q4. Write a query to update the salary by 10%.

Query: db.students.updateMany({}, { \$mul: { salary: 1.1 } }) Output: {

```
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 6,  
  modifiedCount: 6,  
  upsertedCount: 0  
}
```

Q5. Write a query to display all the documents in ascending and descending order of age. Query:

```
db.students.find().sort({ age: 1 })
```

Output: [{

```
  _id: ObjectId('697611febf3c95f1a01e2622'),
  roll_no: 2,
  name: 'Alex', age:
  19,
  salary: 366.74,
  city: 'B',
  phone_no: 456
```

```
},
```

```
{
```

```
  _id: ObjectId('697611febf3c95f1a01e2621'),
  roll_no: 1,
  name: 'Sam',
  age: 20,
  salary: 601.37000000000001,
  city: 'A',
  phone_no: 123
```

```
},
```

```
{
```

```
  _id: ObjectId('697611febf3c95f1a01e2626'),
  roll_no: 6,
  name: 'Danny',
  age: 28,
  salary: 600.93,
  city: 'F',
  phone_no: 892
```

```
},
```

```
{
```

```
  _id: ObjectId('697611febf3c95f1a01e2625'),
```

```
roll_no: 5, name:
'Mukesh', age: 30,
salary: 270.16,
city: 'E',
phone_no: 567
},
{
  _id: ObjectId('697611febf3c95f1a01e2623'),
  roll_no: 3,
  name: 'John',
  age: 40,
  salary: 733.3700000000001,
  city: 'C',
  phone_no: 122
},
{
  _id: ObjectId('697611febf3c95f1a01e2624'),
  roll_no: 4,
  name: 'Bob', age:
  55,
  salary: 746.24,
  city: 'D',
  phone_no: 444
}]
```

Query: db.students.find().sort({ age: -1 }) Output:

```
[
  {
    _id: ObjectId('697611febf3c95f1a01e2624'),
    roll_no: 4,
```

```
name: 'Bob', age:
55,
salary: 746.24,
city: 'D',
phone_no: 444
},
{
  _id: ObjectId('697611febf3c95f1a01e2623'),
  roll_no: 3,
  name: 'John',
  age: 40,
  salary: 733.3700000000001,
  city: 'C',
  phone_no: 122
},
{
  _id: ObjectId('697611febf3c95f1a01e2625'),
  roll_no: 5,
  name: 'Mukesh', age:
30,
  salary: 270.16,
  city: 'E',
  phone_no: 567
},
{
  _id: ObjectId('697611febf3c95f1a01e2626'),
  roll_no: 6,
  name: 'Danny',
  age: 28,
  salary: 600.93,
  city: 'F',
```

```

    phone_no: 892
  },
  {
    _id: ObjectId('697611febf3c95f1a01e2621'),
    roll_no: 1,
    name: 'Sam',
    age: 20,
    salary: 601.3700000000001,
    city: 'A',
    phone_no: 123
  },
  {
    _id: ObjectId('697611febf3c95f1a01e2622'),
    roll_no: 2,
    name: 'Alex', age:
    19,
    salary: 366.74,
    city: 'B',
    phone_no: 456
  }
]

```

Q6. Write a query to display all the documents with City → A,B,C. Query:

```
db.students.find({city: {$in: ["A", "B", "C"]}})
```

Output: [{

```

    _id: ObjectId('697611febf3c95f1a01e2621'),
    roll_no: 1,
    name: 'Sam',
    age: 20,
    salary: 601.3700000000001,
    city: 'A',
    phone_no: 123
  },

```

```
{
  _id: ObjectId('697611febf3c95f1a01e2622'),
  roll_no: 2,
  name: 'Alex', age:
  19,
  salary: 366.74,
  city: 'B',
  phone_no: 456
},
{
  _id: ObjectId('697611febf3c95f1a01e2623'),
  roll_no: 3,
  name: 'John',
  age: 40,
  salary: 733.3700000000001,
  city: 'C',
  phone_no: 122
}
]
```

Q7. Write a query to display only two documents from the entire collection. Query:

```
db.students.find().limit(2)
```

Output: [{

```
  _id: ObjectId('697611febf3c95f1a01e2621'),
  roll_no: 1,
  name: 'Sam',
  age: 20,
  salary: 601.3700000000001,
  city: 'A',
  phone_no: 123
},
{
```

```
_id: ObjectId('697611febf3c95f1a01e2622'),
roll_no: 2,
name: 'Alex', age:
19,
salary: 366.74,
city: 'B',
phone_no: 456
}}
```

Q8. Write a query to delete a document with ROLL_NO:5. Query:

```
db.students.deleteOne({ roll_no: 5})
```

Output: { acknowledged: true, deletedCount: 1 }

Q9. Write a query to display all the documents with AGE greater than 20. Query:

```
db.students.find({ age: { $gt: 20 } })
```

Output: [{

```
_id: ObjectId('697611febf3c95f1a01e2623'),
roll_no: 3,
name: 'John',
age: 40,
salary: 733.3700000000001,
city: 'C',
phone_no: 122
},
{
```

```
_id: ObjectId('697611febf3c95f1a01e2624'),
roll_no: 4,
name: 'Bob', age:
55,
salary: 746.24,
city: 'D',
phone_no: 444
},
```



```
{
  _id: ObjectId('697611febf3c95f1a01e2626'),
  roll_no: 6,
  name: 'Danny',
  age: 28,
  salary: 600.93,
  city: 'F',
  phone_no: 892
}
```

Q10. Write a query to display all the documents with AGE less than 20. Query:

```
db.students.find({ age: { $lt: 20 } })
```

Output: [{

```
  _id: ObjectId('697611febf3c95f1a01e2622'),
  roll_no: 2,
  name: 'Alex', age:
  19,
  salary: 366.74,
  city: 'B',
  phone_no: 456
}]
```

Q11. Write a query to display all the documents with AGE equals to 20. Query:

```
db.students.find({ age: 20 })
```

Output: [{

```
  _id: ObjectId('697611febf3c95f1a01e2621'),
  roll_no: 1,
  name: 'Sam',
  age: 20,
  salary: 601.37000000000001,
  city: 'A',
  phone_no: 123
}]
```

Q12. Write a query to display all the documents with AGE not equals to 20. Query:

```
db.students.find({ age: { $ne: 20 } })
```

Output: [

```
{
  _id: ObjectId('697611febf3c95f1a01e2622'),
  roll_no: 2,
  name: 'Alex', age:
  19,
  salary: 366.74,
  city: 'B',
  phone_no: 456
},
{
  _id: ObjectId('697611febf3c95f1a01e2623'),
  roll_no: 3,
  name: 'John',
  age: 40,
  salary: 733.3700000000001,
  city: 'C',
  phone_no: 122
},
{
  _id: ObjectId('697611febf3c95f1a01e2624'),
  roll_no: 4,
  name: 'Bob', age:
  55,
  salary: 746.24,
  city: 'D',
  phone_no: 444
},
{
```

```
_id: ObjectId('697611febf3c95f1a01e2626'),
roll_no: 6,
name: 'Danny',
age: 28,
salary: 600.93,
city: 'F',
phone_no: 892
}
]
```

Q13. Write a query to display all the documents where AGE is greater than equals to 30. Query:

```
db.students.find({ age: { $gte: 30 } })
```

Output: [

```
{
  _id: ObjectId('697611febf3c95f1a01e2623'),
  roll_no: 3,
  name: 'John',
  age: 40,
  salary: 733.3700000000001,
  city: 'C',
  phone_no: 122
},
{
  _id: ObjectId('697611febf3c95f1a01e2624'),
  roll_no: 4,
  name: 'Bob', age:
  55,
  salary: 746.24,
  city: 'D',
  phone_no: 444
}
]
```

`_id: ObjectId('697611febf3c95f1a01e2621')`
`roll_no: 1`
`name: "Sam"`
`age: 20`
`salary: 601.37000000000001`
`city: "A"`
`phone_no: 123`

`_id: ObjectId('697611febf3c95f1a01e2622')`
`roll_no: 2`
`name: "Alex"`
`age: 19`
`salary: 366.74`
`city: "B"`
`phone_no: 456`

`_id: ObjectId('697611febf3c95f1a01e2623')`
`roll_no: 3`
`name: "John"`
`age: 40`
`salary: 733.37000000000001`
`city: "C"`
`phone_no: 122`

```
_id: ObjectId('697611febf3c95f1a01e2624')
roll_no: 4
name: "Bob"
age: 55
salary: 746.24
city: "D"
phone_no: 444
```

```
_id: ObjectId('697611febf3c95f1a01e2626')
roll_no: 6
name: "Danny"
age: 28
salary: 600.93
city: "F"
phone_no: 892
```